RadeonShift AI

Accelerating the MI300X Hardware Migration via Generative AI

The Problem: CUDA Lock-in

Enterprise AI workloads are facing massive bottlenecks due to a reliance on legacy NVIDIA CUDA codebases.

- Manually migrating a 50,000-line CUDA codebase to ROCm takes **6+ months** of specialized engineering time.
- Portability tools exist (e.g. `hipify-perl`), but they lack the architectural awareness to optimize code for AMD Instinct architecture (like 64-wide wavefronts).
- Teams lack visibility into whether their migrated code will actually compile and run on AMD hardware without direct bare-metal access.

The Solution: RadeonShift AI

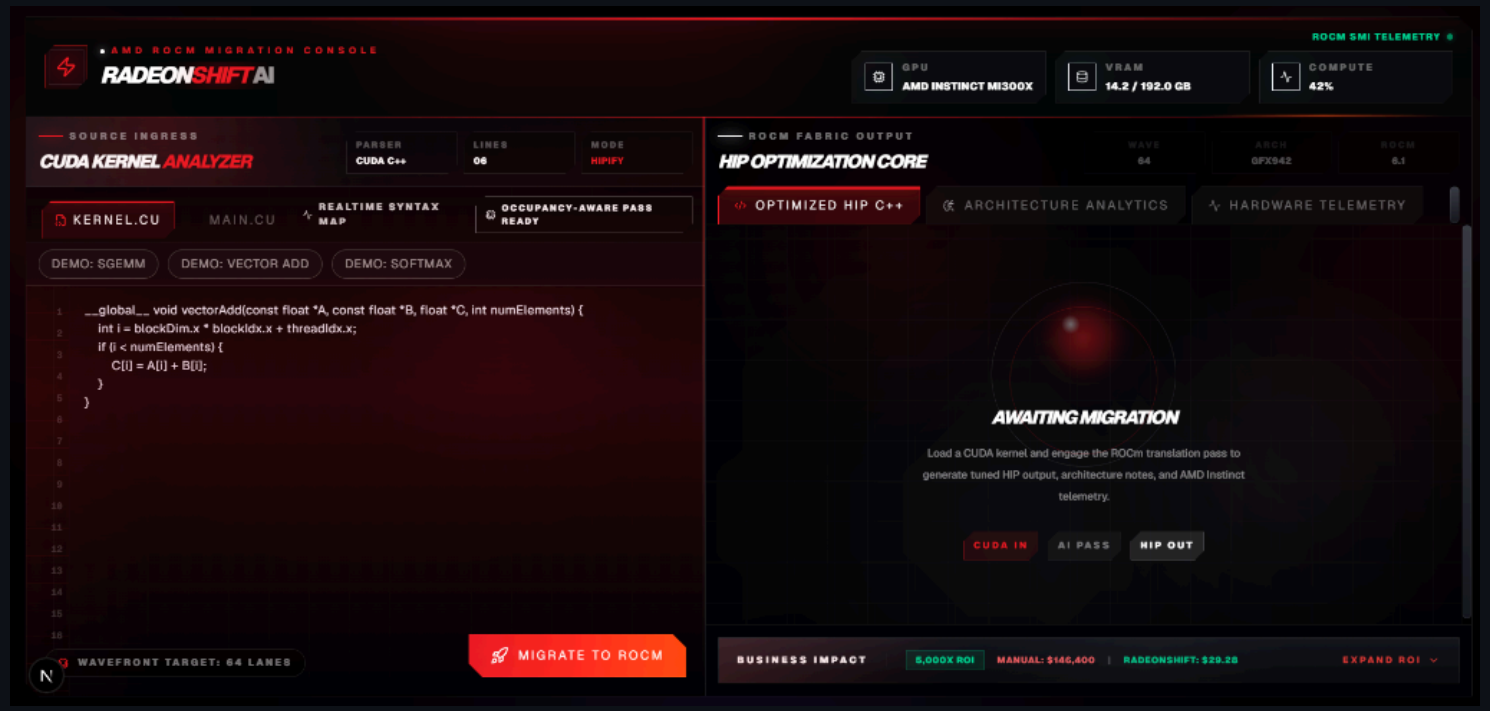
RadeonShift AI is an AI-assisted CUDA kernel migration assistant that translates CUDA to HIP/ROCm and audits architecture-specific migration risks.

1. **Syntax Translation:** Uses Fireworks-hosted translation model for AI Translation.
2. **MoA Audit:** Leverages DeepSeek-V4 to scan for PTX risks, hardcoded warp sizes, and AMD-specific optimizations.
3. **Optional Hardware Evidence:** When the ROCm backend is online, RadeonShift can compile-check generated HIP and show MI300X telemetry. Benchmark mode uses trusted internal kernels and clearly labels live, cached, or unavailable evidence.

Architecture

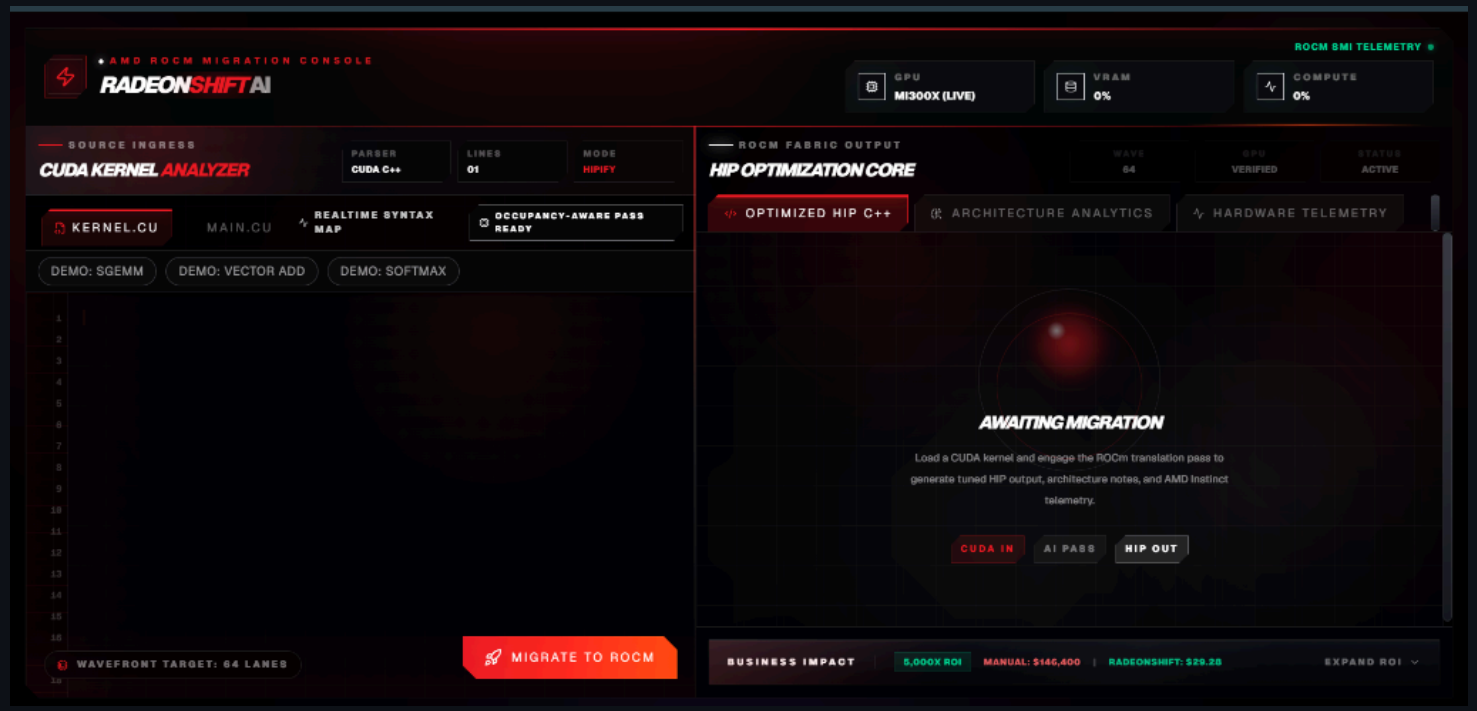
1. **Frontend (Next.js):** Manages user state and renders the dynamic dashboard.
2. **AI Translation Layer (Vercel / Next.js API Routes):** Orchestrates AI translation and Mixture-of-Agents audit via Fireworks AI, operating independently of hardware availability.
3. **Optional Hardware Engine (FastAPI via Pinggy):** A bare-metal ROCm layer that can compile-check generated HIP via `hipcc`, poll `rocm-smi` for telemetry, and run trusted benchmark kernels when connected.

Demo: The RadeonShift Dashboard



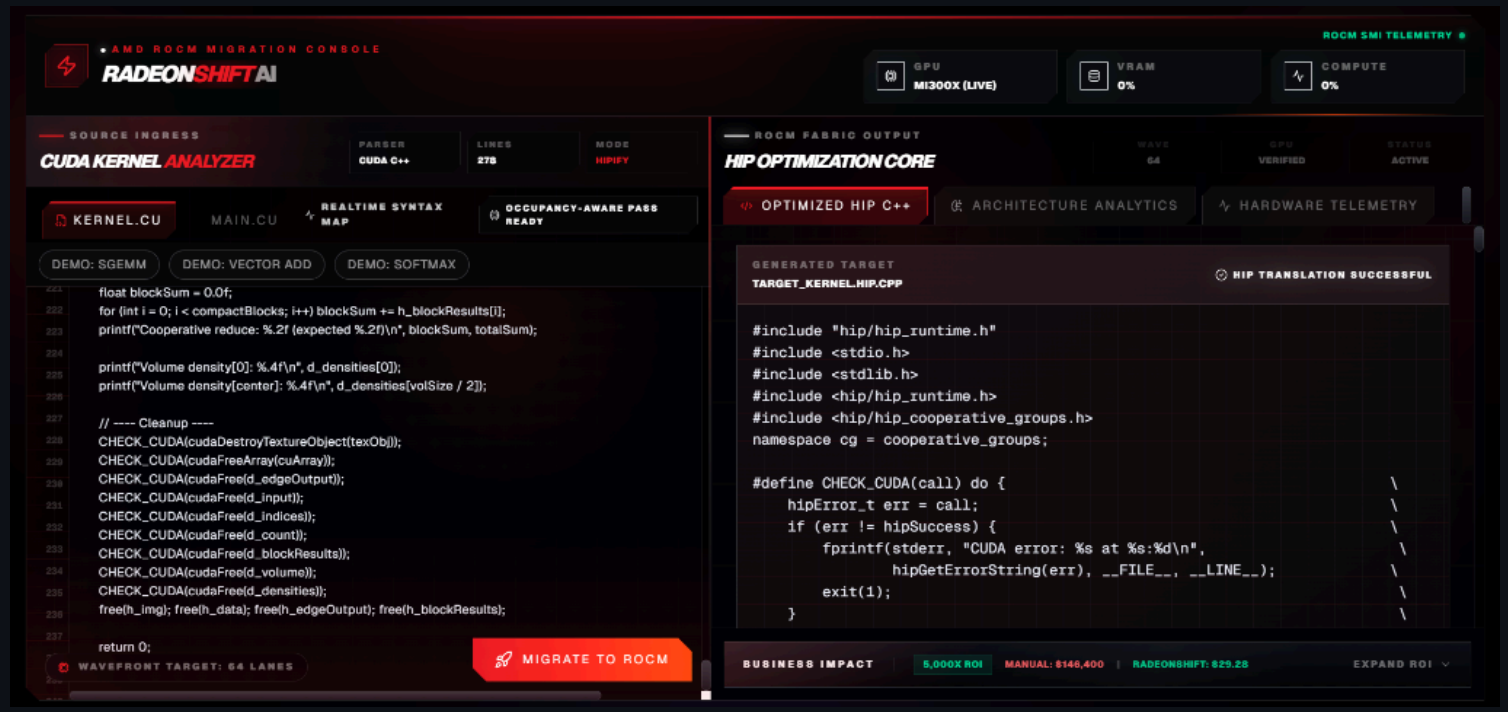
- **Source Code Editor:** Paste raw CUDA C++ kernels.
- **HIP Optimization Core:** View the translated target kernel.
- **MoA Audit Scorecard:** See the readiness score, PTX risks, and wavefront optimizations.

Step 1: The Awaiting Migration State



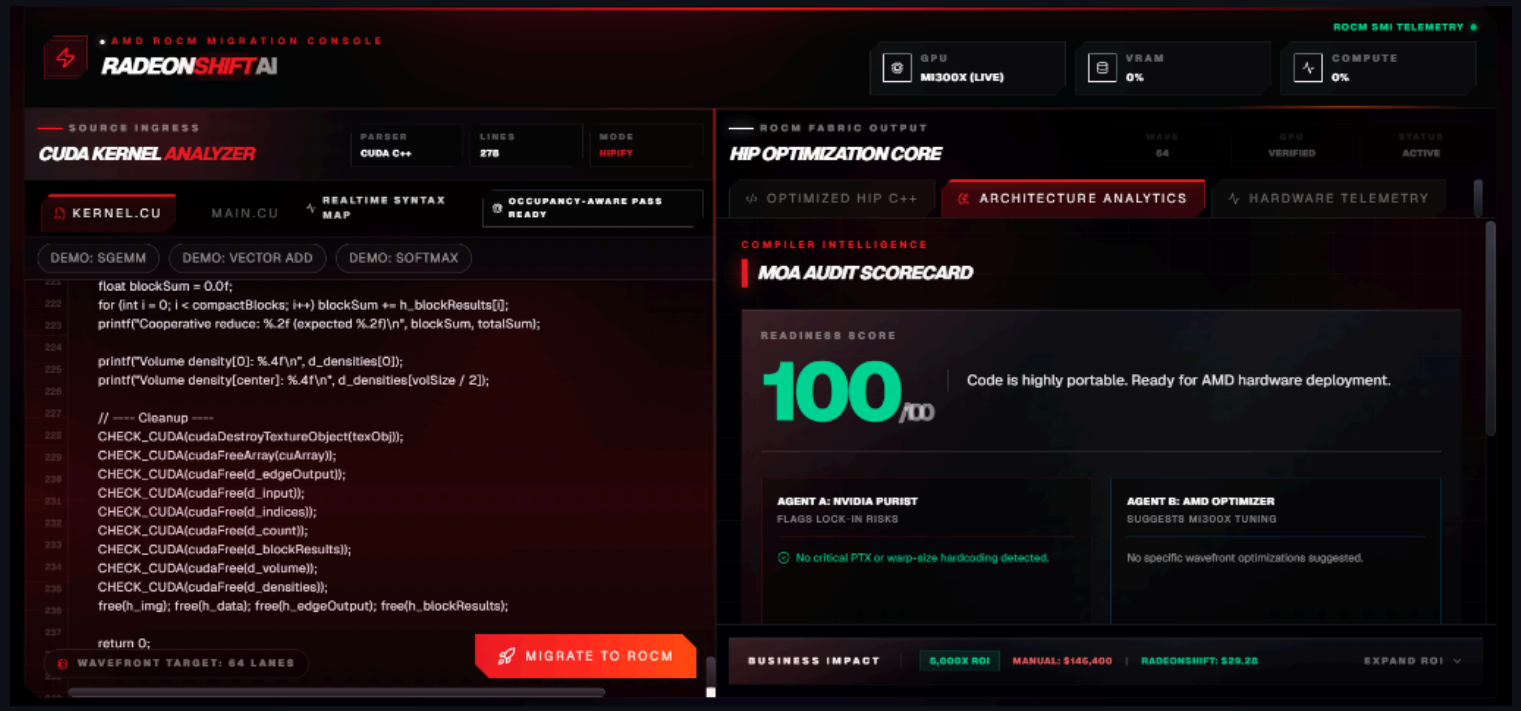
Before translation begins, the platform awaits the CUDA payload. The user inputs their native NVIDIA code into the **CUDA Kernel Analyzer**. The system prepares for the AI-assisted hardware pass.

Step 2: HIP Optimization Core



Upon engaging the ROCm translation pass, the core converts the syntax. The resulting C++ HIP code (`target_kernel.hip.cpp`) is immediately presented, demonstrating generated HIP translation.

Step 3: Architecture Analytics



The MoA Audit Scorecard evaluates the code and computes a readiness score from actual findings. Clean examples can score high, while risky kernels are lowered or capped when PTX, wavefront, WMMA, or async-copy risks appear.

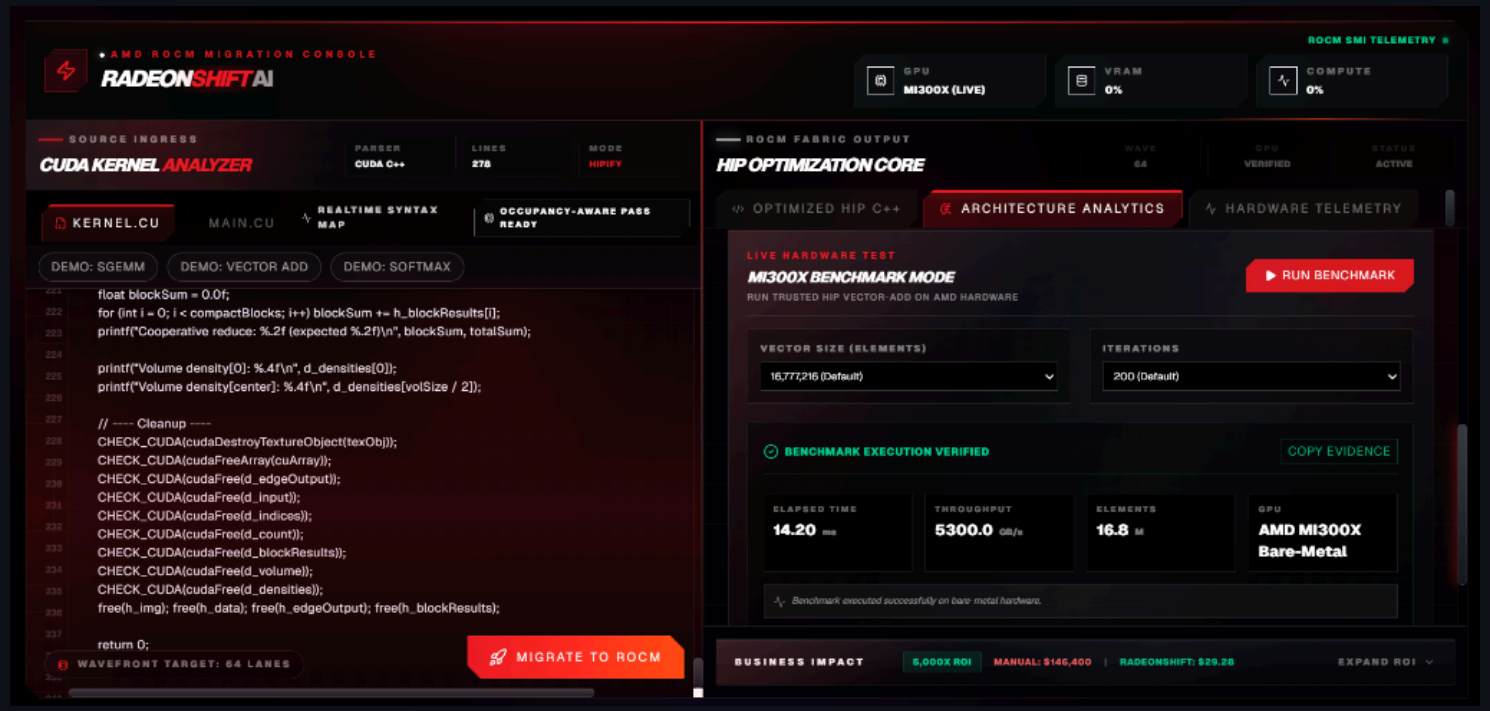
Step 3a: Deterministic Redesign Guardrails

Advanced CUDA Kernel Detection

Not all CUDA code maps 1-to-1. RadeonShift's **Deterministic Rules Engine** catches unsupported architectures (e.g., CUDA WMMA, cooperative groups async-copy).

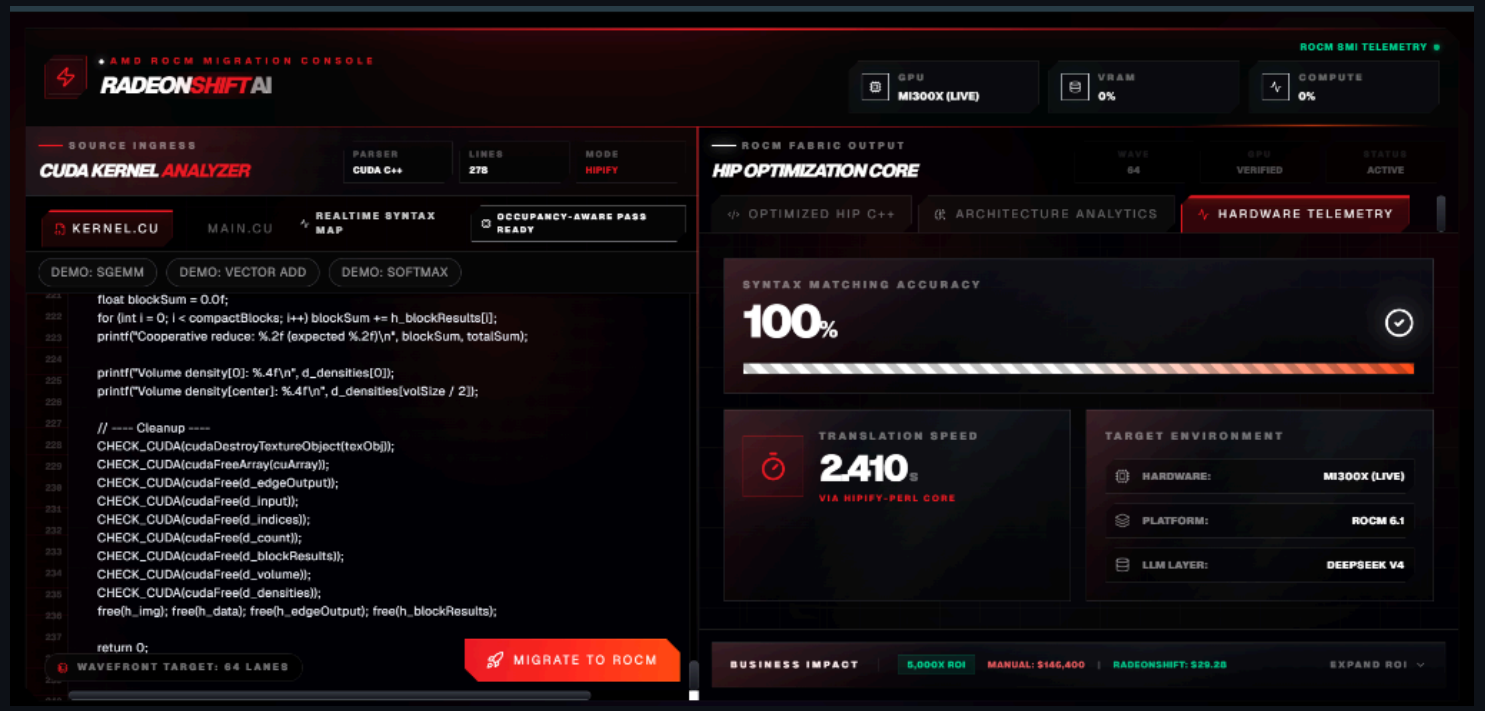
- **Correctness over completeness:** Safely enforces `MANUAL REDESIGN REQUIRED` if code relies on hardware-specific features.
- **Score Capping:** Drops readiness score to < 50% automatically, discouraging unsafe architecture conversions from being treated as ready.

Step 4: Live Hardware Telemetry



When the optional backend is online, the platform connects to a remote bare-metal AMD MI300X environment and surfaces live ROCm telemetry plus compile-check evidence. When offline, hardware fields are explicitly labeled unavailable or cached.

Step 5: Bare-Metal Benchmark Execution



Finally, benchmark mode runs trusted HIP benchmark kernels on MI300X when hardware is online. RadeonShift does not automatically execute arbitrary uploaded kernels; it separates AI audit, compile-check evidence, and runtime benchmark provenance.

Business Value & ROI

Illustrative ROI Model

- **Manual Migration:** 244 Kernels × 4 hrs = 976 Engineer-Hours (~$146,000, based on $150/hr senior GPU engineer rate)
- **RadeonShift Migration:** 244 Kernels × illustrative ~$0.12 AI/compute cost = ~$29 first-pass triage estimate
- **Time Savings:** Initial audit and translation can shrink from weeks of first-pass review to minutes, with human validation still required for production.
- **TAM:** $4.2B illustrative GPU migration and porting services market estimate.

The AMD Infrastructure Story

- **Hardware Target:** Optimized for AMD Instinct MI300X
- **Software Stack:** Built on ROCm 6.x APIs and `hipcc`
- **AI Acceleration:** MoA pipeline runs through the Fireworks-hosted inference API
- **Honest Fallbacks:** If the backend lacks ROCm hardware, the platform gracefully enters AI-Only mode, explicitly labeling any cached evidence without fabricating live metrics

Engineering Retrospective & Challenges

Bridging a Serverless Frontend with Bare-Metal Hardware

1. **Live Telemetry:** Defeated Next.js caching via `force-dynamic` to stream real-time MI300X metrics.
2. **Defensive APIs:** Built robust frontend parsing to gracefully handle sudden JSON schema changes.
3. **CORS Routing:** Bypassed strict mixed-content blocks by tunneling through Pinggy and Next.js `rewrites()`.
4. **Transparent Debugging:** Overhauled error handling to surface remote Python stack traces in the UI.
5. **Structured AI Prompts:** Constrained LLM outputs to raw HIP code or JSON audit findings, reducing UI parsing failures.

Closing & Team

RadeonShift AI is bridging the gap between legacy NVIDIA codebases and the future of AMD compute.

- **GitHub Repository:** shashankh3/RadeonShift-AI
- **Live Demo:** radeon-shift-ai.vercel.app

Thank You!

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